ELECTRICAL ENGINEERING • POWER SYSTEMS SERIES

16:9 CANVAS DECK

# Electrical Transformers Masterclass

A Comprehensive Handbook on Classification, Construction, Operating Principles & Mathematical Formulations

CORE THEORY

### Electromagnetic Principles

Faraday's Law of Mutual Induction, Lenz's Law, magnetic flux linkage, and ideal vs practical transformer operations.

TAXONOMY

### Complete Classification

Power, Distribution, Autotransformers, Instrument Transformers (CT & PT), Isolation, Dry-Type vs Oil-Immersed.

ENGINEERING MATH

### Governing Formulas

EMF equations, transformation ratio, core and copper losses, voltage regulation, equivalent circuit models & efficiency conditions.

# 02. Fundamental Working Principle

FARADAY & LENZ

Electromagnetic induction without any physical moving parts (Static AC Machine)

## Law of Mutual Induction

When alternating voltage $V_1$ is applied to the primary winding, it produces an alternating magnetizing current that establishes a mutual alternating flux $\Phi$ in the ferromagnetic core.

This common flux links with both the primary and secondary windings, inducing an electromotive force (EMF) in each winding according to Faraday's law:

FARADAY'S & LENZ'S LAW

$$e = -N \frac{d\Phi}{dt}$$

Negative sign indicates induced EMF opposes the cause producing it (Lenz's Law).

## Ideal vs. Practical Transformer

- **Zero Winding Resistance ($R_1 = R_2 = 0$):** No copper ($I^2R$) losses.
- **Zero Leakage Flux ($\Phi_{l1} = \Phi_{l2} = 0$):** 100% of flux is confined in the core.
- **Infinite Permeability ($\mu_r \to \infty$):** Zero reluctance; infinitesimal current required to establish flux.
- **Zero Core Losses ($P_h = P_e = 0$):** No hysteresis or eddy current dissipation.

FREQUENCY INVARIANCE PRINCIPLE

$$f_{primary} = f_{secondary} = f_{core}$$

A transformer transforms voltage and current, but NEVER alters electrical frequency.

# 03. Governing Equations of Transformation

CORE FORMULAS

Turn ratio, EMF derivation, and impedance reflection

### THE MASTER EMF EQUATION

$$E_{rms} = 4.44 \cdot f \cdot N \cdot \Phi_m$$

$$E_{rms} = 4.44 \cdot f \cdot N \cdot B_m \cdot A_{core}$$

Where $f$ = Supply Frequency (Hz), $N$ = Number of turns, $\Phi_m$ = Peak magnetic flux (Weber), $B_m$ = Maximum flux density ($\text{Tesla}$ or $\text{Wb/m}^2$), $A$ = Net core cross-sectional area ($\text{m}^2$).

### TRANSFORMATION & TURNS RATIO ($K$)

$$\frac{V_2}{V_1} \approx \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2} = k$$

If $k > 1 \implies$ Step-Up; If $k < 1 \implies$ Step-Down. Note current transforms inversely to voltage.

### TURNS RATIO ($A$) & IMPEDANCE REFLECTION

$$a = \frac{N_1}{N_2} = \frac{1}{k}$$

Impedance transferred from Secondary to Primary:

$$R_2' = a^2 \cdot R_2 = \left(\frac{N_1}{N_2}\right)^2 R_2$$

$$X_2' = a^2 \cdot X_2 = \left(\frac{N_1}{N_2}\right)^2 X_2$$

$$Z_2' = a^2 \cdot Z_2$$

### EMF PER TURN (UNIVERSAL INVARIANT)

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44 \cdot f \cdot \Phi_m$$

Both windings always experience the exact same induced voltage per turn.

# 04. Comprehensive Transformer Taxonomy

CLASSIFICATION

Categorization by application, structural construction, phase count, and insulation

BY APPLICATION

## Grid Purpose

- **Power:** High voltage transmission (>200 MVA).
- **Distribution:** End-user supply (≤2500 kVA, 11kV/415V).
- **Autotransformer:** Inter-grid voltage tie.
- **Instrument:** CT & PT metering/protection.
- **Isolation:** 1:1 safety & harmonic decoupling.

BY CONSTRUCTION

## Magnetic Design

- **Core Type:** Windings encircle core limbs; preferred for high voltage.
- **Shell Type:** Core encircles windings; excellent mechanical bracing.
- **Berry Type:** Multi-limb radial shell; high cooling surface.
- **Toroidal:** Doughnut core; near-zero leakage flux.

BY COOLING MEDIUM

## Thermal Class

- **Dry-Type (VPI/Cast Resin):** Fire-safe for indoor commercial facilities.
- **Oil-Immersed (ONAN/ONAF):** Mineral or synthetic ester oil for high thermal dissipation.
- **Forced Cooling (OFAF/ODAF):** Pumped oil & fans for massive power units.

BY PHASE & FREQ

## System Service

- **Single-Phase (1-Φ):** Residential pole-top, traction.
- **Three-Phase (3-Φ):** Standard commercial & grid power.
- **High-Frequency (Ferrite):** SMPS, flyback, inverter stages (kHz to MHz).
- **Earthing/Grounding:** Zig-zag winding creates neutral.

# 05. Core Construction & Magnetic Properties

METALLURGY & PHYSICS

Metallurgy, silicon steel laminations, and suppression of parasitic losses

### Cold Rolled Grain Oriented (CRGO) Steel

Modern transformer cores use silicon alloyed steel (approx 3.2% Si) cold-rolled to align crystal grains in the rolling direction, yielding very high permeability $\mu_r$ and narrow hysteresis loop.

- **Lamination Thickness:** Typically **0.23 mm to 0.35 mm** insulated with Carlite varnish.
- **Saturation Flux Density ($B_{sat}$):** Operates between **1.6 T and 1.8 T** to prevent saturation under overvoltage.
- **Step-Lap Mitred Joints:** 45° mitered overlapping joints reduce joint reluctance and inrush acoustics.

### Amorphous Metal (Metglas) Cores

Produced by ultra-rapid cooling (1,000,000 °C/sec) of molten alloy, resulting in non-crystalline glass-like atomic structure.

- **No-Load Loss Reduction:** Reduces core hysteresis & eddy losses by up to **70% – 80%** compared to standard CRGO.
- **Trade-Offs:** Lower saturation flux density ($B_{sat} \approx 1.56\text{ T}$), mechanically brittle, higher initial cost.
- **Target Application:** High-efficiency green distribution transformers.

STACKING FACTOR ($K_I$)

$$A_{net} = k_i \cdot A_{gross} \quad (k_i \approx 0.90 \text{ to } 0.96)$$

Accounts for insulation varnish occupying physical core volume.

# 06. Windings & Dielectric Insulation Systems

INSULATION & THERMAL

Conductors, winding geometries, and thermal classification standards

## Winding Types & Placement

Low-voltage (LV) winding is placed nearest to the core limb to reduce required dielectric clearance to ground. High-voltage (HV) winding surrounds the LV winding.

- **Cylindrical / Layer Winding:** Low voltage, up to 33 kV.
- **Helical Winding:** High current LV windings (interleaved multi-strip).
- **Continuous Disc Winding:** Standard for medium & high voltage systems; excellent axial short-circuit withstand.
- **Foil Winding:** Copper or aluminum sheet across full axial length, ideal for Cast Resin Dry Transformers.

## Thermal Insulation Classes (IEC / NEMA)

| CLASS | MAX TEMP | TYPICAL MATERIALS |
|---|---|---|
| **Class A** | 105 °C | Kraft paper, pressboard in mineral oil. |
| **Class B** | 130 °C | Inorganic materials with bonding varnish. |
| **Class F** | 155 °C | Resin-encapsulated glass cloth, mica. |
| **Class H** | 180 °C | Silicone elastomers, Nomex aramid paper. |
| **Class C** | > 180 °C | Porcelain, ceramic, high-temp composites. |

# 07. Power vs. Distribution Transformers

GRID ENGINEERING

Operating philosophy, load cycles, and efficiency optimization points

| DESIGN FEATURE | POWER TRANSFORMER | DISTRIBUTION TRANSFORMER |
|---|---|---|
| **Grid Location** | Generating stations & transmission substations (>132 kV). | Distribution networks, commercial facilities, end-users (11kV/415V). |
| **Load Profile** | Continuous near 100% full load around the clock. | Widely fluctuating load (average 50% to 70% load over 24 hours). |
| **Efficiency Target** | Engineered for **Maximum Commercial Efficiency at 100% Full Load** ($P_i = P_{cu,FL}$). | Engineered for **Maximum All-Day Efficiency at 50% – 70% Load** ($P_i \ll P_{cu,FL}$). |
| **Impedance (%Z)** | High (10% to 15%) to limit catastrophic downstream fault levels. | Low (4% to 5%) to maintain tight voltage regulation at consumer taps. |

**ALL-DAY ENERGY EFFICIENCY ($\ETA_{ALL-DAY}$)**

$$\eta_{all-day} = \frac{\text{Energy Output (kWh in 24 hours)}}{\text{Energy Input (kWh in 24 hours)}} = \frac{\text{Output kWh}}{\text{Output kWh} + 24 \cdot P_i + \sum (x_t^2 P_{cu} \cdot t)} \times 100\%$$

Core loss ($P_i$) runs 24/7 regardless of load; copper loss scales with square of hourly load factor ($x_t^2$).

# 08. Autotransformers & Specialty Units

SPECIALTY MACHINERY

Common winding configurations, copper economy, and isolation decoupling

## Autotransformer: Principle & Economy

Utilizes a single continuous winding shared between primary and secondary, where energy is transferred partly by **conduction** and partly by **induction**.

COPPER WEIGHT SAVING

$$\text{Weight of Cu}_{Auto} = (1 - k) \cdot \text{Weight of Cu}_{2-Winding}$$

Maximum economy when $k \to 1$ (e.g. stepping 400 kV to 220 kV).

POWER CONDUCTION VS INDUCTION

$$S_{transformed} = (1 - k) S_{total} \quad \Big| \quad S_{conducted} = k \cdot S_{total}$$

## Specialty Transformers

- **Galvanic Isolation Transformer (1:1 Ratio):** Zero physical connection between primary and secondary. Decouples ground loops, eliminates common-mode surges, protects sensitive hospital/medical equipment.
- **Toroidal Transformer:** Circular donut core with no air gap. Virtually zero acoustic hum, extremely low stray magnetic field (ideal for audio amplifiers and precision lab instrumentation).
- **Grounding / Earthing Transformer (Zig-Zag):** Provides a stable artificial neutral for ungrounded Delta networks to drain fault currents without circulating zero-sequence harmonics.

# 09. Instrument Transformers: CT & PT

PROTECTION & METERING

Precision measurement, relay protection, and the critical secondary open-circuit rule

CURRENT TRANSFORMER (CT)

## Series Measuring Instrument

Steps down massive primary currents (e.g. 1000A) to standard secondary values (**1A or 5A**) for protective relays and energy meters.

**CRITICAL SAFETY RULE:**

NEVER open-circuit a live CT secondary! With zero secondary counter-MMF, total primary MMF saturates the core, inducing lethal kilovolts across terminals and causing explosive dielectric failure.

POTENTIAL TRANSFORMER (PT)

## Parallel Voltage Instrument

Steps down high distribution/transmission voltages (e.g. 11kV, 33kV) to safe standardized secondary levels (typically **110V or 110V/$\sqrt{3}$**).

- **Burden Rating:** Apparent power (VA) drawn by connected relays/meters.
- **Accuracy Classes:**
  - **Class 0.2S / 0.5:** Revenue billing meters.
  - **Class 5P10 / 5P20:** Protective relaying (5% composite error up to 10x or 20x rated current).

# 10. Cooling Codes & Substation Protection

SAFETY & THERMAL

Thermal classification (IEC 60076) and transformer auxiliary safety equipment

## Standard Cooling Classifications

| CODE | INTERNAL FLUID | EXTERNAL CIRCULATION |
|---|---|---|
| **ONAN** | Oil Natural convection | Air Natural radiation/fins. Standard ≤ 5 MVA. |
| **ONAF** | Oil Natural convection | Air Forced with motorized cooling fans. |
| **OFAF** | Oil Forced with pumps | Air Forced cooling radiator banks. |
| **ODAF** | Oil Directed to windings | Air Forced radiator banks (>100 MVA). |
| **AN / AF** | Air Natural convection | Air Forced fans (Cast Resin Dry-Type). |

## Core Protective Devices

- **Buchholz Relay:** Dual-float gas-actuated relay located between main tank and conservator. Triggers alarm on minor gas accumulation; trips breaker on violent oil surges.
- **Silica Gel Breather:** Dehydrates air entering conservator during thermal breathing (turns from blue to pink when saturated).
- **Pressure Relief Valve (PRV):** Spring-loaded rapid rupture device relieving internal overpressure within milliseconds.
- **WTI / OTI:** Winding & Oil Temperature Indicators with trip/alarm contacts.

# 11. Equivalent Circuit & Fault Impedance

CIRCUIT ANALYSIS

Mathematical parameter referral to primary winding & short circuit calculation

## Exact Equivalent Circuit Parameters

- **Series Branch:** Winding resistance ($R_1, R_2$) and leakage reactance ($X_1, X_2$).
- **Parallel Shunt Branch:** Core-loss conductance $R_c$ (iron losses) and magnetizing susceptance $X_m$ (creates flux $\Phi_m$).

REFERRED TO PRIMARY SIDE ($A = N_1/N_2$)

$$R_{01} = R_1 + R_2' = R_1 + a^2 R_2$$

$$X_{01} = X_1 + X_2' = X_1 + a^2 X_2$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

## Percentage Impedance (%Z) & Fault Current

The percentage of rated primary voltage required to circulate full-load rated current under short circuit.

PERCENTAGE IMPEDANCE (%Z)

$$\%Z = \frac{I_{FL} \cdot Z_{01}}{V_1} \times 100\%$$

SHORT CIRCUIT SYMMETRICAL FAULT CURRENT ($I_{SC}$)

$$I_{sc} = \frac{I_{FL}}{\%Z / 100} = I_{FL} \times \left(\frac{100}{\%Z}\right)$$

If $\%Z = 5\%$, short circuit current is **20x rated full-load current**!

# 12. Losses, Efficiency & Maximum Derivation

EFFICIENCY ANALYSIS

Iron vs copper losses and mathematical condition for peak efficiency

## Loss Components

1. CORE LOSS (CONSTANT WITH LOAD)

$$P_i = P_{hysteresis} + P_{eddy}$$

Hysteresis: $P_h = k_h f B_m^{1.6}$ | Eddy: $P_e = k_e f^2 B_m^2 t^2$

2. COPPER LOSS (VARIABLE WITH LOAD FACTOR $X$)

$$P_{cu}(x) = x^2 \cdot P_{cu,FL} = x^2 \cdot I_{2,FL}^2 R_{02}$$

At 50% load ($x = 0.5$), copper losses drop to **25%** of full load!

## Efficiency Formulation & Peak Condition

COMMERCIAL EFFICIENCY EQUATION

$$\eta = \frac{x \cdot S \cdot \cos\phi}{x \cdot S \cdot \cos\phi + P_i + x^2 P_{cu,FL}} \times 100\%$$

Where $S$ = Rated kVA, $\cos\phi$ = Load power factor.

CONDITION FOR MAXIMUM EFFICIENCY ($\FRAC{D\ETA}{DX} = 0$)

$$\text{Variable Copper Loss} = \text{Constant Iron Loss}$$

$$x^2 \cdot P_{cu,FL} = P_i \implies x = \sqrt{\frac{P_i}{P_{cu,FL}}}$$

Load for max $\eta$: $\text{kVA}_{\max\eta} = \text{Rated kVA} \times \sqrt{\frac{P_i}{P_{cu,FL}}}$

# 13. Voltage Regulation & Testing Procedures

BENCH TESTING & VR

Formulas for regulation and standard OC/SC bench testing

## Voltage Regulation (VR)

The fractional change in secondary terminal voltage from no-load to full-load expressed as percentage of full-load voltage.

APPROXIMATE REGULATION EQUATION

$$VR \approx \frac{I_2 (R_{02}\cos\phi \pm X_{02}\sin\phi)}{V_{2,FL}} \times 100\%$$

**+** sign for **Lagging PF** (Inductive load $\implies$ voltage drops)
**–** sign for **Leading PF** (Capacitive load $\implies$ voltage rises - Ferranti effect)

ZERO VOLTAGE REGULATION CONDITION (LEADING PF)

$$\tan\phi = -\frac{R_{02}}{X_{02}} \implies \cos\phi = \frac{X_{02}}{Z_{02}} \text{ (leading)}$$

## Factory Testing Benchmarks

- **Open-Circuit (OC) Test:** Conducted on **Low-Voltage (LV) side** at rated voltage with HV side open. Wattmeter measures pure **Iron Losses ($P_i$)**. Determines shunt parameters: $R_c$ and $X_m$.
- **Short-Circuit (SC) Test:** Conducted on **High-Voltage (HV) side** at rated current with LV side shorted. Voltage applied is only 5% to 10% of rated $V$. Wattmeter measures pure **Full-Load Copper Losses ($P_{cu,FL}$)**. Determines series parameters: $R_{01}$, $X_{01}$, and $Z_{01}$.

# 14. Three-Phase Systems & Master Formula Sheet

CHEATSHEET SUMMARY

Vector groups, 3-Φ power formulas, and universal quick reference

## Three-Phase Configurations & Vector Groups

| CONNECTION | PRIMARY / SEC | STANDARD APPLICATION |
|---|---|---|
| **Dyn11** | Delta / Star neutral | Standard Indian/European distribution (LV neutral). |
| **Yd1** | Star / Delta | Generator step-up transmission unit. |
| **Yy0** | Star / Star | High-voltage tie lines with tertiary delta. |
| **Dd0** | Delta / Delta | Heavy low-voltage industrial plant systems. |

THREE-PHASE APPARENT & ACTIVE POWER

$$S = \sqrt{3} \cdot V_L \cdot I_L \quad (\text{kVA})$$
$$P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\phi \quad (\text{kW})$$

## Universal Engineering Cheatsheet

- **EMF:** $E = 4.44 \cdot f \cdot N \cdot B_m \cdot A$
- **Ratio:** $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = a$
- **Referred R:** $R_{01} = R_1 + a^2 R_2$
- **Max $\eta$ Load:** $kVA_{\max} = kVA_{FL} \times \sqrt{P_i / P_{cu,FL}}$
- **Fault Current:** $I_{sc} = I_{FL} / (\%Z / 100)$
- **All-Day $\eta$:** $\sum \text{kWh}_{out} / (\sum \text{kWh}_{out} + 24P_i + \sum x^2 P_{cu})$
- **Autotransformer Cu Saving:** $\text{Saving} = k \times \text{Cu}_{2\text{-winding}}$